ADVERSE SELECTION IN THE ANNUITY MARKET – MATHEMATICAL MODELLING AND EMPIRICAL TESTING

DOTE 3011 Behavioural Economics and Business Decisions

Presented By
Group 3 – Austin NG Long Ching (1155159226)

FLOW OF PROPOSAL

Literature Review

- Review on the main assumptions and mathematical arguments on how individuals respond to the heterogeneous early health signals

Research Design and Process

- Model Modification: Relax the setting of perfect rationality of people to non-perfect rationality, involve present bias, extend the stage of the model
- Data Collection: Collect data using questionnaire
- Empirical Model Building: Try to build multiple regression model to explain adverse selection in the annuity market

Contribution and Future Direction

- Explain the contribution of this research and provide some possible directions for future research

LITERATURE REVIEW

- Lau, S.-H. P. and Zhang, Q. (2023), Heterogeneous early health signals and deferred annuities, Working Paper (main paper)

- Agar Brugiavini (1993), Uncertainty resolution and the timing of annuity purchases, Journal of Public Economics 50, p.31-62 (supplementary paper)

LITERATURE REVIEW

- ## Main Ideas of the Paper by Lau and Zhang (2023)

- **Adverse selection is the main reason why the purchase of annuity in the market is low.**

- **The provision of deferred annuities with the assumption that annuitants have heterogeneous health signals proves that the immediate annuity market is incompletely crowded out.**

- Lau and Zhang (2023) used the method of **backward induction**. They assume that an annuitant's life is divided into three periods (while no exact duration of period is provided), from period 0 to period 2, and determine the consumptions with the following steps:

    1. Annuitants consume anything they own in period 2, assuming NO bequest motive

    2. Based on the consumption choice in period 2, annuitants make optimal consumption choice in period 1

    3. Based on the consumption choice in period 1, annuitants make optimal consumption choice in period 0

LITERATURE REVIEW

- Building Blocks of the Model

Perfect Rationality	*Adverse Selection	Uniform Pricing
**Survival Uncertainty	***Two Types of Annuity	

* Annuitants are assumed to have private health information. In the revised model, annuitants will also be assumed that they have private information regarding the willingness to form a family and whether they are inclined to present awards instead of future payoff

** Annuitants are certain that they can be alive in period 0 and period 1 but not in period 2

*** They are deferred annuity (buy in period 0 only) and immediate annuity (buy in period 1 only)

LITERATURE REVIEW

Mathematical Formulation

- It follows the standard procedure in expected utility maximization. That is, in **period 1**:

$$
\begin{aligned}
U_{z\theta} &= u(c^1_{z\theta}) + \frac{\theta}{1+\rho} u(c^2_{z\theta}) \\
&= u\left((1+r)w - \frac{q}{1+r}\beta_z - \frac{p}{1+r}\alpha_{z\theta}\right) + \frac{\theta}{1+\rho} u(\beta_z + \alpha_{z\theta})
\end{aligned}
$$

where

- $u(\cdot)$ is the utility function, $u'(\cdot) > 0$ and $u''(\cdot) < 0$ (law of diminishing marginal utility)
- w is the wealth level
- θ is the survival probability in period 2, and θ follows a probability density function
- ρ is the subjective discount rate,
- z is the health signal,
- β_z = amount of deferred annuity purchased in period 0,
- $\alpha_{z\theta}$ = amount of immediate annuity purchased in period 1 (θ is updated in period 1),
- r is risk-free interest rate,
- p and q are the respective prices of immediate and deferred annuity

LITERATURE REVIEW

- **Mathematical Formulation**

Given the annuitant knows the optimal consumption choice he or she will make in period 1, the annuitant makes the purchase decisions to optimize the expected utility in **period 0**. That is,

$$E\left(U_{z\theta}|z\right) = \int_{\underline{\theta}}^{\overline{\theta}} U_{z\theta} f_{\theta|z}\left(\theta|z\right) d\theta,$$

RESEARCH DESIGN – MODEL MODIFICATION

- Extension of The Period of the Model

  - According to Lau and Zhang (2023), they followed the work from Brugiavini (1993) and defined period 0 as working stage, period 1 as early retirement stage and period 2 as later retirement stage

  - While this definition brings mathematical convenience, this model is **restrictive** and may be unable to truly describe the adulthood life cycle of an annuitant

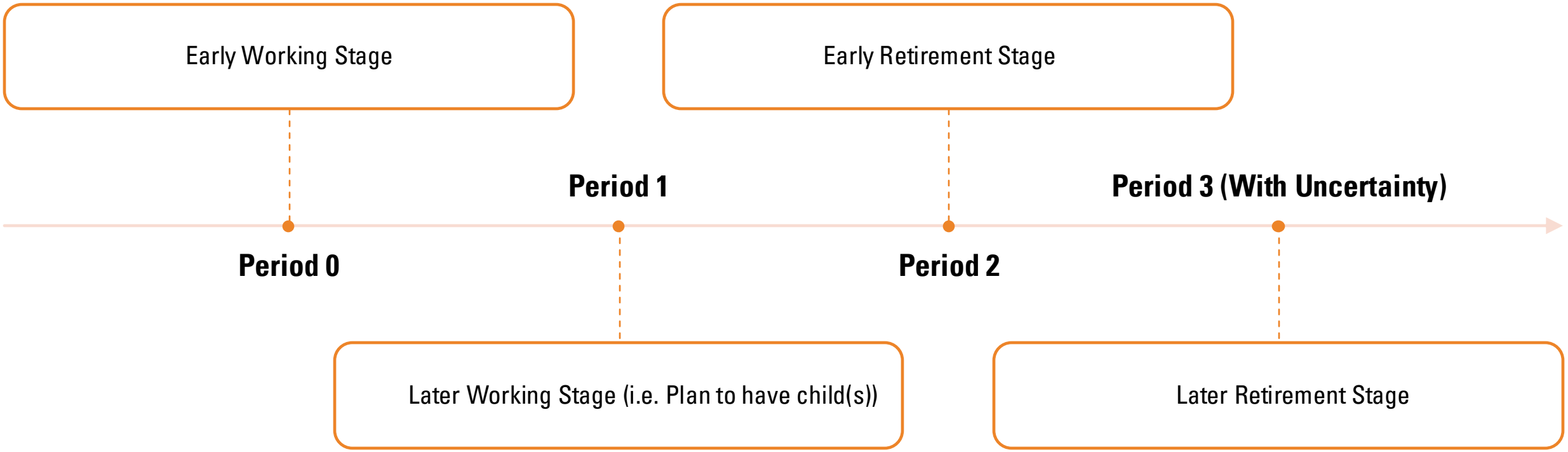
  - If an annuitant plans to have child(s) in the future, they may spend less amount of money in buying annuities as additional expenses on education of their child(s) occur, or even quit the market

  - This is an example of **Adverse Selection**

  - We can extend the 3-stage model to a 4-stage model by adding one period as later working stage or working stage (with child)

RESEARCH DESIGN – MODEL MODIFICATION

- Revised Life Cycle Timeline



RESEARCH DESIGN – MODEL MODIFICATION

- **Relaxation of Perfect Rationality Assumption**

- In this literature's setting, annuitants are always assumed to be perfectly rational. However, this is not always the case

- Why?

- The annuitant may have a certain level of present bias, that is, he or she places more values on the present rewards than the future rewards

- This is also an example of **Adverse Selection** – people with stronger present bias may be less willing to buy annuities or quit the annuity market, but this is unknown by the annuity providers

RESEARCH DESIGN – MODEL MODIFICATION

- Using Hyperbolic Discounting

- Recall in the research paper that the authors use 1/(1+ρ) as the discounting factor. This is called **exponential discounting**

$$
\begin{aligned}
U_{z\theta} &= u(c^1_{z\theta}) + \frac{\theta}{1+\rho} u(c^2_{z\theta}) \\
&= u\left((1+r)w - \frac{q}{1+r}\beta_z - \frac{p}{1+r}\alpha_{z\theta}\right) + \frac{\theta}{1+\rho} u(\beta_z + \alpha_{z\theta})
\end{aligned}
$$

- To revise the mathematical formulation, we may use **hyperbolic discounting** instead with the revised life cycle timeline. In **period 1**, it will become

$$
U_{z\theta} = u(c^1_{z\theta}) + \beta[(\frac{1}{1+\rho})u(c^2_{z\theta}) + \theta(\frac{1}{1+\rho})^2 u(c^3_{z\theta})]
$$

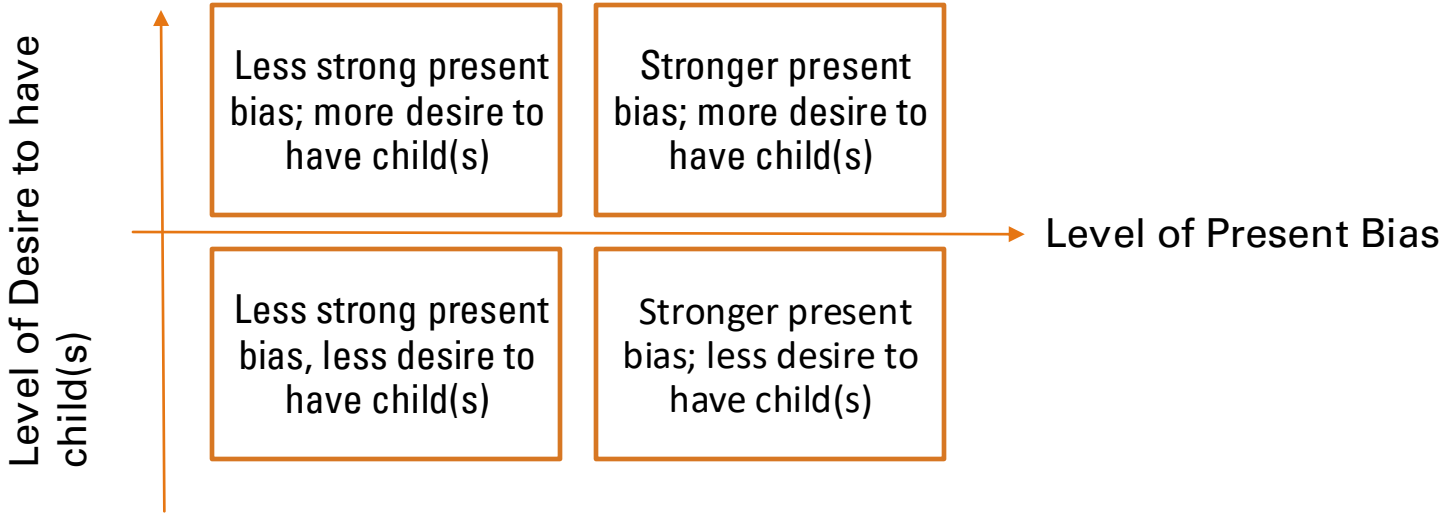
RESEARCH DESIGN – DATA COLLECTION

- **Testing Adverse Selection in the Annuity Market**

- Recall the three-stage model is extended to a four-stage model (with the inclusion of a stage that people plan to have their child(s)), and the present bias is also added to the model

- Our goal is to test **whether the improved model can be better at explaining the adverse selection problem in the annuity market**

- Types of annuity purchase data needed for empirical testing:



RESEARCH DESIGN – DATA COLLECTION

- **Methods of Collecting Data**

- Proposed Questions for the Survey (for participants aged 18 or above only):

1. What is your age?

○ 18 to 29 ○ 29 to 40 ○ 40 or Above

2. How many years have you been working?

○ 0 to 5 Years ○ 5 to 10 Years ○ 10 to 15 Years ○ More than 15 Years

3. Have you been buying any annuities?

○ Yes ○ No

4. If the answer in question 3 is yes, what is the premium of annuity per year?

○ $0 - $2000 ○ $2000 - $4000 ○ $4000 - $6000 ○ $6000 or Above

RESEARCH DESIGN – DATA COLLECTION

- Methods of Collecting Data

5. Rate your level of desire to have child (s) in 5 to 10 years after.

○ Unlikely	○ Slightly Unlikely	○ Likely	○ Highly Likely

6. How much money do you expect a least to get 31 days after if you sacrifice getting $1000 tomorrow? (tick one box only)

○ $1030	○ $1060	○ $1090	○ $1120
○ $1150	○ $1180	○ $1210	○ $1240
○ $1270	○ $1300	○ $1340	○ $1370
○ $1400	○ $1430	○ $1460	○ $1490 or Above

RESEARCH DESIGN – DATA COLLECTION

- Methods of Collecting Data

7. How much money do you expect at least to get 381 days after if you sacrifice getting $1000 351 days after? (tick one box only)

○ $1030	○ $1060	○ $1090	○ $1120
○ $1150	○ $1180	○ $1210	○ $1240
○ $1270	○ $1300	○ $1340	○ $1370
○ $1400	○ $1430	○ $1460	○ $1490 or Above

RESEARCH DESIGN – EMPIRICAL MODEL BUILDING

- **Ordinary Least Square Regression**

  - Regress the annuity premium (Premium) on the level of desire to have child(s) (L_c), where 1 = Unlikely, 2 = Slightly Unlikely, 3 = Likely, 4 = Highly likely) and the level of present bias (L_b), and the interaction terms LcLb (level of desire to have child(s) x level of present bias)

  - That is, Premium = $\alpha + \gamma_1 L_c + \gamma_2 L_b + \gamma_3 LcLb + \varepsilon$, where α is the intercept term and ε is the error term

- **Expected Result**

  - $\gamma_1 < 0$ (∂ Premium / ∂ Level of desire to have child(s) < 0)

  - Explanation: If people have higher level of desire to have child(s), they may spend less money to buy annuities in order to save for their child(s)

RESEARCH DESIGN – EMPIRICAL MODEL BUILDING

- **Expected Result (continued)**

- $\gamma_2 < 0$ (∂ Premium / ∂ Level of present bias < 0)
- Explanation: If people have higher level of present bias, they may prefer the immediate enjoyment more, thus they may be less willing to buy annuities at this moment

- $\gamma_3 < 0$ (∂ Premium / ∂ {Level of present bias x Level of desire to have child(s)} < 0)
- Explanation: As both level of present bias and level of desire to have child(s) are expected to have negative effects on the annuity premium, we can expect that the combined effect is also negative

CONTRIBUTION AND FUTURE RESEARCH DIRECTION

- **Contribution of This Research**

  - This research tries to explain the existence of adverse selection in the annuity market by relaxing the assumption that annuitants are perfectly rational, and annuitants can also have private information regarding to the willingness to have child(s) in the future and present bias.

  - It also explains the quantitative effect of those factors on the premium of annuity, i.e., the amount annuitants pay for buying the annuities (per year) by a multiple linear regression

  - This modification is also expected to increase the explanatory power of the theoretical model constructed by Lau and Zhang (2023)

- **Future Research Direction**

  - It is important to consider further improving the accuracy of the model, i.e., if omitted variable bias occur, more advanced causal inference techniques like instrumental variable regression / two-stage least square regression may have to be used